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JANUARY 9, 2020 BY **LISA CLANCY**

How to write a figure caption

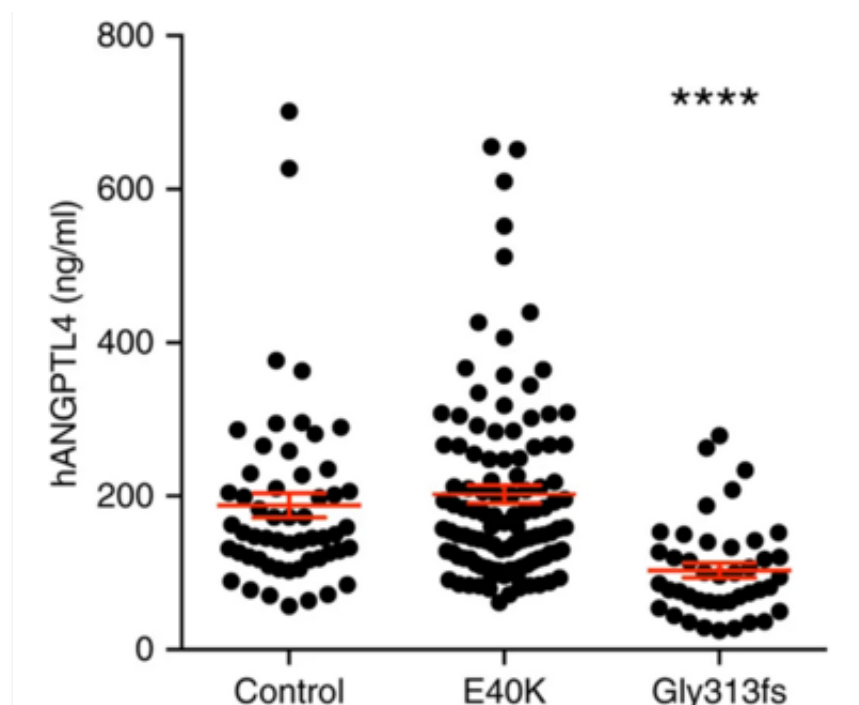
Figure captions should be standalone, i.e., descriptive enough to be understood without having to refer to the main text. Effective captions typically include the following elements:

a **declarative title** that summarises the result or major finding of the data you are presenting in the figure;

a brief description of the **methods** necessary to understand the figure without having to refer to the main text;

statistical information, for example, number of replicates, asterisks denoting P-values, statistical tests, etc.

Example:



Declarative title →

Methods →

Statistical information →

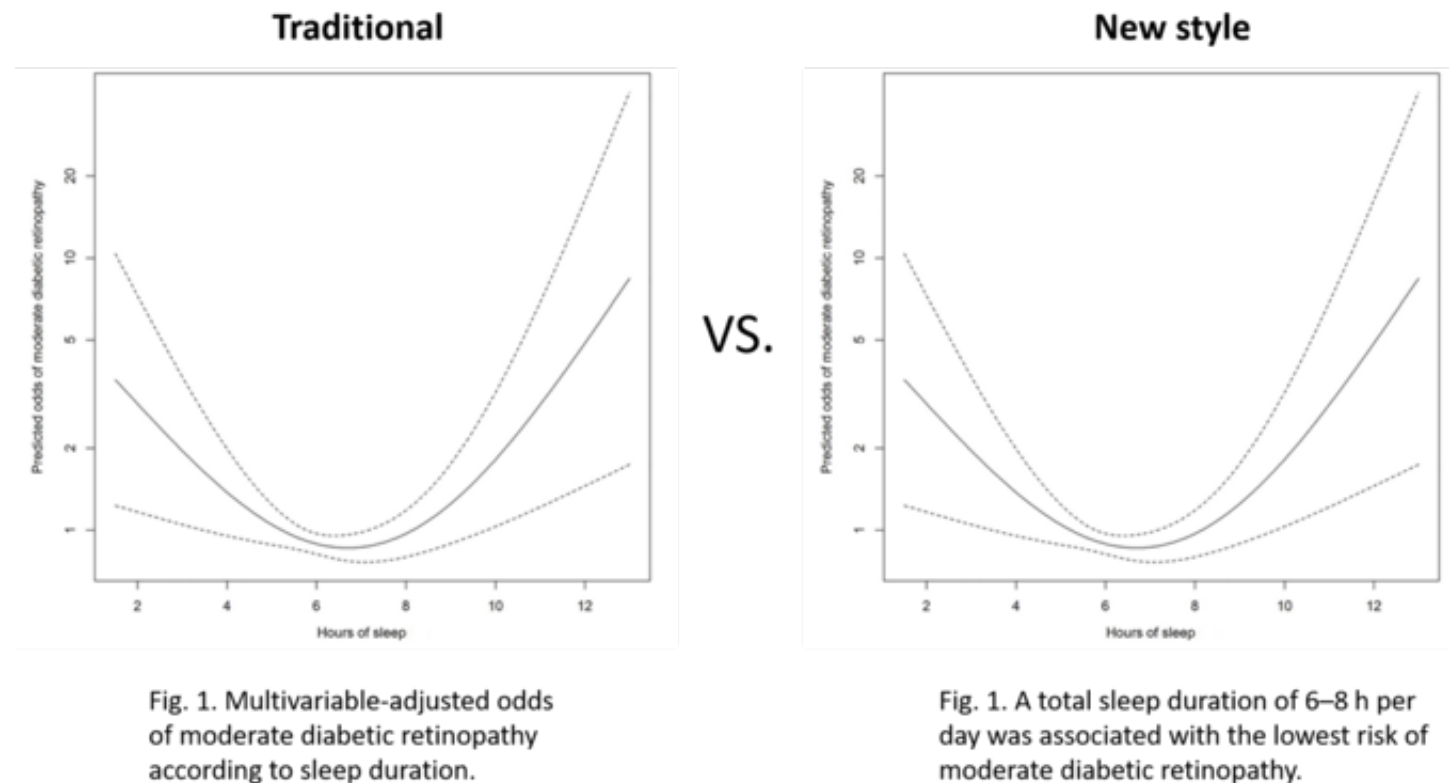
Plasma ANGPTL4 levels were reduced in p.G313fs carriers. ANGPTL4 plasma levels were measured in fasted serum from 86 heterozygous p.E40K, 42 heterozygous p.G313fs variant carriers, and 55 controls matched for age, sex, and body mass index. Statistics performed by unpaired t-test with Welch's correction, comparing each variant carriers group to controls, **** $p < 0.0001$

[1].

A declarative title – will that bias the reader?

Traditionally, captions did not overtly state the main finding of the data being presented. Instead, a typical figure caption would only indicate the variables being discussed, for

example, “Scatterplot showing the relationship between X and Y”. The onus was on the reader to draw their own conclusions from the figure. However, there has been a shift away from this style of caption in recent years (see below for a direct comparison of the different styles).



[2].

This new style of figure caption offers a number of benefits. It helps the reader quickly grasp the author’s main take-home message. This should, in theory, entice the reader to

read more of the text, i.e., if the reader quickly understands the main finding of the figure, they are likely to want to move on and build on the new knowledge. Writing in this style should also help focus the author at the construction stage, urging them to create figures that communicate their point most effectively. Finally, highly focused captions may prevent “data-dumps”, in which the author “dumps” data they cannot make sense of in the hope that someone else can [3].

You may be concerned, however, that a declarative title is not objective, i.e., it may bias the reader. However, it is likely you have already tailored your paper to present evidence for/against a hypothesis. Thus, in many ways, the whole paper is (acceptably) biased. Why state a conclusion overtly in the main text but hide it in the figures [3]?

Caption or legend?

Ask yourself if there are elements in the figure caption that would be better served being included in a legend. In many cases, as with the following example, describing the figure elements in a legend is preferable.

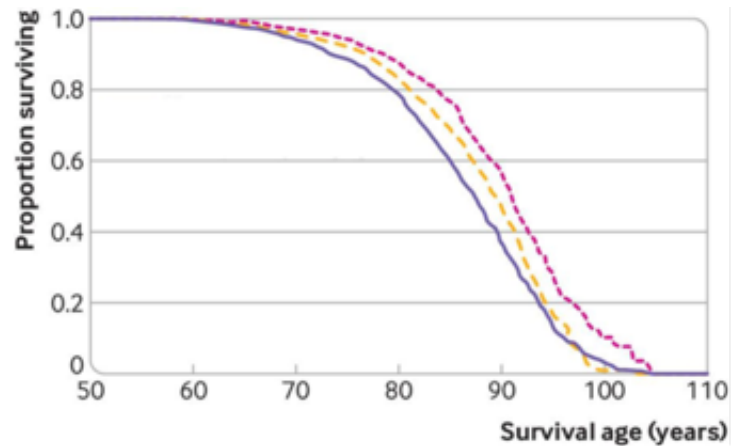


Fig. 1. Arts engagement had a protective association with longevity in older adults. Adjusted for demographic, socioeconomic, health related, behavioural, and social confounding factors. Solid blue line represents adults who never engaged with arts activities; yellow dashed line represents those who infrequently engaged with arts activities; pink dashed line represents those who frequently engaged with arts activities.

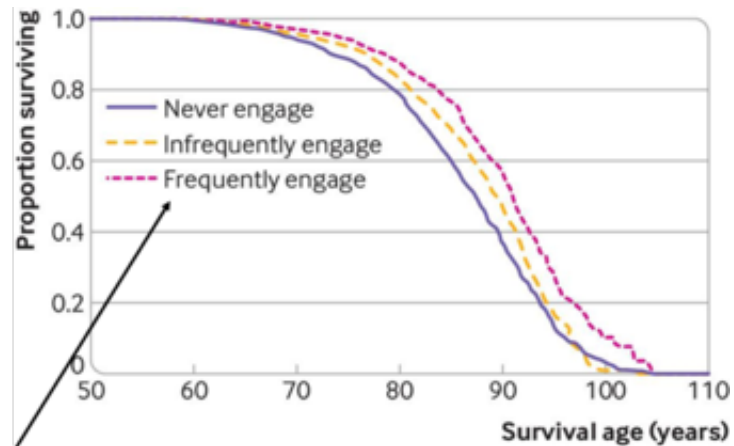
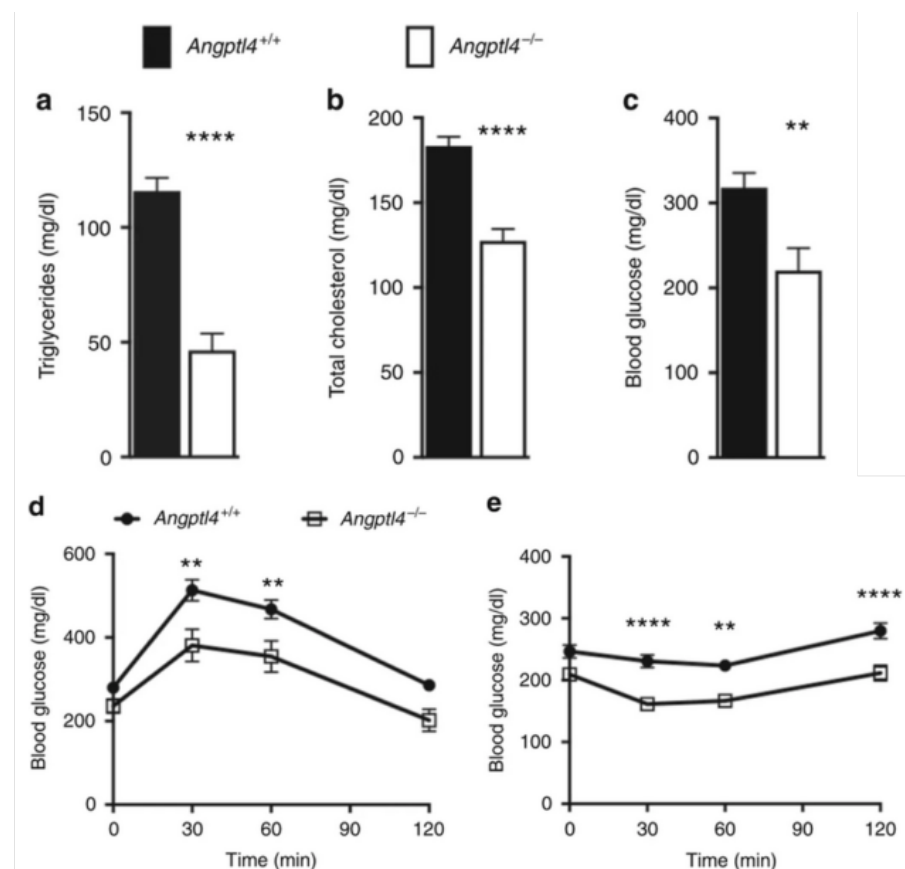


Fig. 1. Arts engagement had a protective association with longevity in older adults. Adjusted for demographic, socioeconomic, health related, behavioural, and social confounding factors.

[4].

Scope of a figure

Ideally, each figure should answer only one question. You may have performed multiple independent experiments to answer one research question. The results of these independent experiments can be combined into a multi-panel figure, with one overarching declarative title.



Overarching
declarative title

Figure 2. *Angptl4*^{-/-} mice had improved glucose homeostasis. (a) Serum triglycerides, (b) total cholesterol, and (c) blood glucose levels in *Angptl4*^{-/-} and littermate control mice on a high-fat diet for 9 weeks. (d) Oral glucose tolerance test and (e) insulin tolerance test in the animals described in (a–c). All groups had 9–11 animals. Values are mean ± SEM. Statistical analysis by Welch's t-test (a) and 2-way ANOVA with Sidak's post-test (d, e), ***p* < 0.001, *****p* < 0.0001. The study was conducted in three different cohorts of mice, with qualitatively similar results in each replicate.

[1].

References

1. Gusarova V, O'Dushlaine C, Teslovich TM, Benotti PN, Mirshahi T, Gottesman O, Van Hout CV, Murray MF, Mahajan A, Nielsen JB, Fritsche L. Genetic inactivation of ANGPTL4 improves glucose homeostasis and is associated with reduced risk of diabetes. *Nature Communications*. 2018 Jun 13;9(1):2252.
2. Tan NY, Chew M, Tham YC, Nguyen QD, Yasuda M, Cheng CY, Wong TY, Sabanayagam C. Associations between sleep duration, sleep quality and diabetic retinopathy. *PloS One*. 2018 May 24;13(5):e0196399.
3. Kroodsma DE. A quick fix for figure legends and table headings. *The Auk*. 2000 Oct 1;117(4):1081-3.
4. Fancourt D, Steptoe A. The art of life and death: 14 year follow-up analyses of associations between arts engagement and mortality in the English Longitudinal Study of Ageing. *BMJ*. 2019 Dec 18;367.

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